Block 4 — Problem Solving · Creativity · Gestalt · Impediments

Unit 1: Nature of Problem Solving

Definition

"A higher-order cognitive process requiring modulation and control of more routine skills." **Baron (2001):** "efforts to develop or choose among various responses to attain desired goals." **Witting & Williams III (1984):** "use of thought processes to overcome obstacles." Two processes: **problem orientation** (motivational/attitudinal/affective) + **problem-solving skills** (cognitive-behavioural steps).

Types of Problems

Well-defined: nature + info needed are clear (e.g., math equation, jigsaw). **Ill-defined:** nature unclear, info less obvious (e.g., "how to bring peace").

Greeno (1978) — 3 categories:

Arrangement rearrange elements (anagrams, jigsaw)
Inducing structure identify/construct relationships
Transformation change initial → goal state (Tower of Hanoi); requires **means-end analysis**

Characteristics of Difficult Problems

Intransparency · Commencement opacity · Continuation opacity · **Polytely** (multiple goals) · Inexpressiveness · Opposition · Transience · Complexity · Enumerability · Connectivity · Heterogeneity · Dynamics · Temporal constraints · Temporal sensitivity · Phase effects · Dynamic unpredictability

Unit 2: Stages of Problem Solving

Stages (General)

1. Recognise/define the problem → **2. Represent** the problem → possible solutions → **4. Evaluate** alternatives → **5. Implement & verify** solution

Master Strategies

Algorithm: specific procedure that guarantees solution if followed correctly. Includes **systematic random search** (orderly, keeps record) and **unsystematic random search** (no record, may repeat).
Heuristic: rules of thumb; no guarantee but faster. E.g., familiar letter combinations in anagrams.

Specific Techniques

Technique	Description	Key Names
Generate & Test	Generate solutions, test against criteria. Loses effectiveness with many possibilities.	—
Means-Ends Analysis	Detect difference between current & goal state; reduce it. Involves **subgoals**.	Newell, Shaw & Simon (1958)
Backward Search	Start at goal, work backward. Useful when goal uniquely specified.	Wickelgren (1974)
Planning/Analogy	Split into simple/complex; solve simple first. Analogy = using earlier solution.	—
Thinking Aloud	Concurrent vs retrospective verbalisation. **Protocol** → **problem behaviour graph**.	Ericsson & Simon; Newell (1977)

Schooler, Ohlsson & Brooks (1993): verbalisation interferes with insight problems — non-reportable/unconscious processes.

Other Strategies

Abstraction · Divide & conquer · Hypothesis testing · **Lateral thinking** · Method of focal objects · Reduction · Research · Root cause analysis · Trial-and-error · **Brainstorming** (rules: express all; no idea too wild; quantity; no criticism until done)

Wickelgren's 7 Strategies (State-Action Tree)

(1) Inference · (2) Classification of action sequences (equivalence classes) · (3) State evaluation & hill climbing · (4) Subgoals · (5) Contradiction · (6) Working backward · (7) Finding relations between problems

IDEAL (Bransford & Stein)

Identify → **D**efine → **E**xplore → **A**ct → **L**ook back

Unit 3: Theoretical Approaches

Traditional Approach

Associative learning (classical/instrumental conditioning). S-R associations. Problem difficulty = strength of correct vs incorrect. Extinction weakens wrong; reinforcement strengthens correct. Stresses **transfer of prior learning**.

Gestalt Approach

Kohler (1925, *Mentality of Apes*): reorganisation of objects; flash of **insight**.
Insight: "sudden discovery of correct solution following incorrect attempts based on trial and error."
Wertheimer (1959): Reproductive thinking (rote, tried-and-true) vs **Productive thinking** (insight + creativity; new structuring).
Nine-dot problem: requires restructuring beyond visual boundaries. **Sagacity:** ability to see into situation, discriminate important aspects.
Insight vs non-insight: metacognitive assessments accurate for non-insight, not for insight.

Information Processing & GPS

Newell, Shaw & Simon (1958) — General Problem Solver:
Limited WM · Large LTM · Serial processor · Heuristics (not algorithms).
Problem space = set of nodes (states of knowledge); solvers move via **operators**. **Problem Space Hypothesis:** every state corresponds to mental node.

Newell's Approach

Problems solved by exploring paths. At **knowledge level**, knowledge = "a means to an end, a resource for behaviour." Goal = select one possible action.

Problem Solving as Modelling

Construction of **situation-specific model / case model**. **Domain model** (augmented assumptions) + **Task model** (meaning of goals) → **competency theory**. **Specialised principle of rationality** (basis for "why" questions).
Heidegger (1968): essence of thinking not reducible to computing.

Expert Systems / AI

Knowledge base + inference rules + search engine + user interface. **Problem space hypothesis** used to create expert systems.

Unit 4: Impediments to Problem Solving

5 Barrier Types

Perceptual · **Emotional** · **Intellectual** · **Environmental** · **Cultural**

Functional Fixedness (Duncker, 1935)

Tendency to see objects as having only one typical use. **Candle problem:** 43% solved (boxes as containers) → 100% (boxes empty). **Simmel (1953) coin problem:** 3+3+2 grouping. Cognitive barrier; lack of transfer. Remedy = reorganisation.

Einstellung (Mental Set) — Luchins (1942)

Water jar problems: mechanised approach blocks better solutions. Repetition of successful method → applies to new problems where simpler method exists.

Environmental Blocks (8)

Management style · Distractions · Physical discomfort · Lack of support · Stress · Lack of communication · Monotonous work · Expectations of others

Cultural Blocks (6)

Unquestioning acceptance of status quo · Dislike of change · Fantasy/humour not productive · Feelings/intuition unreliable · Over-emphasis on competition or cooperation · Taboos

Overcoming Blocks

Re-learning for own blocks; sidestep others' blocks. "Fantasy and humour connected by unlikely combination of ideas — innovative solutions arise the same way."

Decision-Making Models

Haynes: 3-step reflective questions for ethical dilemmas. **Hall (2001):** questions for educators — What is the basic issue? What principle is at risk? Who benefits? Benefits > harm?